Reg. No | | | | | | | | | | |

[13AE56]

FIFTH SEMESTER B.E EXAMINATION, DECEMBER – 2015

COMPOSITE MATERIALS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.

1.	**a.**	List the materials used for important structures in a military aircraft along with critical properties required for selection based on design service loads.	**06**
	b.	Explain shear test for composites in two railed system with a neat sketch; how it differs from three railed test.	**08**
2.	**a.**	What is tempered glass and laminated glass? How it is fabricated? What are its properties?	**08**
	b.	Explain three tests that must be conducted for aircraft glass to qualify for applications.	**06**
3.	**a.**	Explain resin transfer molding process of fabricating composites with a neat sketch. What are its applications?	**08**
	b.	Explain manufacturing of carbon fibers from petroleum pitch with a neat schematic sketch.	**06**
4.	**a.**	What is impact damage cone in a composite laminate? Explain the progressive failure mechanisms in a laminated composite.	**08**
	b.	Explain tensile and compression properties of CFRP.	**06**
5.	**a.**	What are the constituents of ablative coating and its functions? Explain the mechanism of ablation.	**06**
	b.	What are the sources of high temperature in aircraft and space vehicles? What type cooling methods can be adopted for dissipation of heat generated? What are the advantages of having a ceramic coating on a metal structure?	**08**
6.	**a.**	Explain diffusion bonding process for manufacturing MMC's with a neat sketch.	**06**
	b.	Explain the properties and applications of AI-SiC and Mg-SiC in aircraft structures.	**08**

SCHOOL OF ENGINEERING & TECHNOLOGY
LIBRARY
DATE
JAIN UNIVERSITY BANGALORE

7.	**a.**	Explain possible defects in honey-comb structures with their effects and possible NDT techniques for inspection.	**06**
	b.	Explain mechanisms of curing epoxy based composites with a curing cycle, by indicating pressure, temperature and viscosity changes.	**08**
8.	**a.**	What is ultrasonic testing? Explain pulse echo and through transmission techniques with neat sketches. List the applications of ultrasonic testing.	**08**
	b.	Explain the following with respect to magnetic particle inspection	**06**
		i. Circular and longitudinal magnetization	
		ii. Coil technique and central conductor technique of magnetization.	
		iii. Direct and indirect magnetization.	